

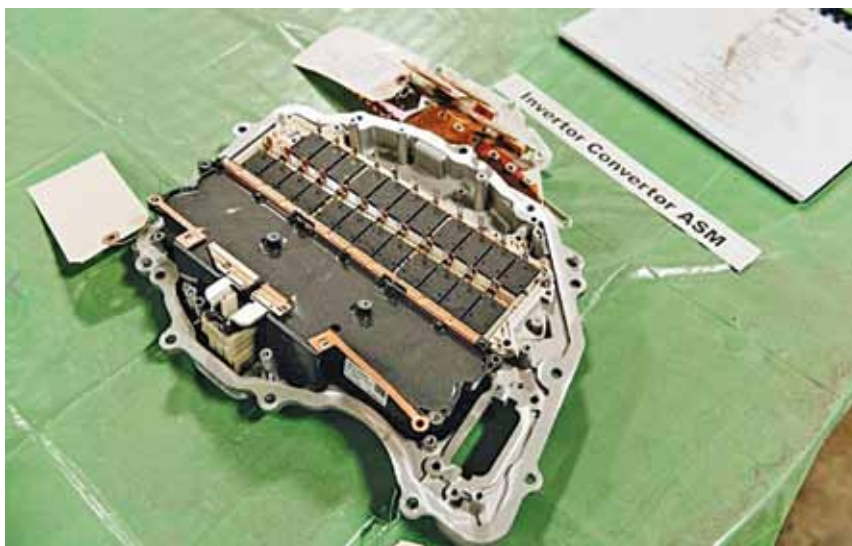
of silicon can increase the driving range per charge, speed up charging time, and increase efficiency by delivering the same range with lesser battery capacity and weight.

Importance of SiC in automobiles

Today, the electric vehicle (EV) sector is growing rapidly and governments all over the world are taking initiatives for the preservation of the environment. Moreover, petrol prices and CO₂ emissions are a matter of great concern. This is why we can see large-scale investments in EVs. Within the next decade, EVs might very well dominate the vehicle market.

But it is important to note that EVs still cannot compete with conventional vehicles in terms of cost—they are significantly expensive. Hence, EV designers and researchers are looking to increase EV efficiency as that will translate to lower costs. Exploring SiCs has pushed EVs towards this goal.

The very properties that SiC has enable it to be used in devices operating at high temperatures and/or high voltages. Automobiles, especially EVs and hybrid EVs, contain many high-voltage systems like the high-voltage battery and its charging sub-systems. This makes SiC a good choice for use in such applications. With SiC technology incorporated in MOSFETs and diodes, the whole system can be made smaller, lighter, and more efficient.



Model 3 main inverter featuring 24 SiC MOSFET modules from ST Microelectronics

Electrical systems in EVs, such as battery management systems (BMS) and onboard chargers (OBC), are crucial since the battery is a fundamental feature of EVs. Typically, the OBC is used to charge the battery using AC mains when the car is parked. This is generally a unidirectional process—from mains to the battery. But OBCs can be used the other way round too, for example, when you want to power your home using the energy stored in your EV battery. By using SiCs in such bidirectional chargers, more efficient bidirectionality can be obtained.

Reasons for the success of SiC

According to BusinessWire, “Global SiC market was valued at US\$850.6 million in 2020, and is estimated to reach US\$2670.25 million in 2027”.

Here are some reasons for its success.

- Its on-state resistance and switching losses are considerably lower
- High thermal conductivity compared to silicon, and hence lesser cooling requirements
- Allows faster heat dissipation from components
- Provides high power density, and hence better efficiency
- Wide bandgap
- Efficient even in harsh environments due to exceptional hardness and strength
- Ability to switch at three to five times higher system frequencies
- High thermal shock resistance and low thermal expansion
- Chemical inertness because chemical bonds in SiC are stronger

SiC and the BFOM Equation (Baliga's Figure of Merit)

In the 1980s, Dr Jayant Baliga, an electrical engineer, professor, and inventor of IGBTs suggested replacing silicon with wide bandgap semiconductors to enhance the performance of power devices. “In the latter half of the 1980s, the technology for the growth of silicon carbide was developed with the culmination of commercial availability of wafers from CREE Research Corporation,” says Baliga in his book titled Silicon Carbide Power Devices.

Dr Baliga's famous equation, commonly referred to as Baliga's Figure of Merit (BFOM), predicts that SiC would be much more efficient than silicon. Baliga was well aware that SiC devices could replace his own invention—the IGBT. “It has been my privilege to assist in the establishment of national programs to fund research on silicon carbide technology in the United States, Japan, and Switzerland-Sweden,” he says.

The 15cm to 20cm transition

Many companies have already adopted SiC technology in conjunction with traditional silicon. Tesla, in 2018, added SiC MOSFETs from STMicroelectronics in its inverter for Model 3, due to which the weight of the inverter was reduced by half as compared to other models. The industry believes that although SiC production is not easy on our pockets, the huge demand in the